
Survey Instrument

The Solutions Dimension

Participant and facilitator forms, with scoring notes

What this instrument is for

It measures three things at once: whether participants found the activity usable and worthwhile, whether the four design requirements from Cycle 1 were actually met, and whether confidence on specific tasks moved. Items marked [†] are carried verbatim from the Cycle 1 survey so the two rounds stay comparable. Items marked ^R are reverse-scored.

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1. Before You Administer This

The one mistake to avoid

Satisfaction is not learning. Cycle 1 reported that participants enjoyed the activity and found it easy, and reviewers correctly objected that this said nothing about whether anyone learned anything. This instrument has the same limit. It can support claims about *usability*, *perceived value*, *workload*, and *confidence*. It cannot support a claim about learning gains — only the performance measures M1–M4 in the study design can do that, and they come from the session artifacts, not from here. Write the claims down before you collect anything, and check afterwards that none of them drifted.

1.1. Length and Timing

The whole participant instrument is 38 items and runs 6–8 minutes. That is already longer than a single play session, so resist adding to it. If you must cut, drop Part F (cognitive load) first and Part D (realism) second; keep Parts B, C and E intact.

When	What	Why then
Before play, 2 min	Part A (background), Part B (pre-confidence)	Confidence must be measured before exposure or the pre/post comparison is worthless
After the debrief, 6 min	Parts C–H	After debrief, not after Phase 3 — the debrief is part of the intervention
Same day	Facilitator form (Part I)	Memory for interventions decays fast

1.2. Linking Pre to Post Without Names

Participants generate their own code, which lets you pair pre and post responses without collecting identity. Put this at the top of both forms:

Please create your code: the first two letters of the street you grew up on, the last two digits of your phone number, and the first two letters of your mother's first name. For example, ELM Street, phone ending 47, mother Rosa gives EL47R0. Use exactly the same code on both surveys.

Expect 10–15% of codes not to match. Report the pairing rate.

1.3. Response Format

All agreement items use a five-point scale: **1 Strongly disagree, 2 Disagree, 3 Neither agree nor disagree, 4 Agree, 5 Strongly agree**. Confidence items use a 0–100 slider. Cognitive load items use 0–10.

Do not offer a “not applicable” option on agreement items; it invites satisficing. Do allow every item to be skipped.

2. Part A — Background

Reviewers of the Cycle 1 paper specifically asked for this. Six items.

- A1. What is your primary field of study? (*Computer science / Information technology / Engineering / Natural sciences / Mathematics / Social sciences / Humanities / Business / Health sciences / Other*)

- A2. What is your current level? (*First or second year undergraduate / Third or fourth year undergraduate / Graduate student / Faculty or staff / Other*)
- A3. How many cybersecurity courses have you completed? (*None / One / Two or three / More than three*)
- A4. Before today, had you heard of the CIS Critical Security Controls? (*No / Heard of them / Used them*)
- A5. How often do you play board or card games with other people? (*Never / A few times a year / Monthly / Weekly or more*)
- A6. In your team today, which best describes you? (*I did most of the talking / I contributed about equally / I mostly listened / Unsure*)

Why A5 and A6 are here

A5 tests a confound worth ruling out: if only experienced gamers enjoyed it, the activity is less useful than it looks. A6 gives you a cheap check on whether non-majors were talked over, which is the failure mode the facilitator manual warns about, and it is more honest than asking the facilitator to judge it.

3. Part B — Confidence, Before Play

Administer *before* the session. Same five items reappear as Part G.

How confident are you that you could do each of the following right now? Slide from 0 (cannot do at all) to 100 (highly certain I can do).

- B1. Explain to a non-technical person why an organization should keep backups.
- B2. Given a described attack, name two security measures that would make it less likely to succeed.
- B3. Choose between two security measures when the organization can only afford one, and say why.
- B4. Explain the difference between making an attack less likely and making it less damaging.
- B5. Tell an organization what risk remains after the measures they have put in place.

Why these five

They are written to Bandura's guidance for self-efficacy scales: specific tasks rather than general traits, and graded confidence rather than agreement. Each maps to something the game actually asks a player to do — B3 is the budget, B4 is the risk matrix, B5 is Phase 3 — so movement is interpretable rather than diffuse.

4. Part C — The Cycle 1 Anchor Items

Carry these four verbatim. They are the only bridge between the two rounds, and rewording them for elegance would break it.

- C1. † The Security Cards helped me learn.
- C2. † The Security Cards are easy to use.
- C3. † The Security Cards raised or maintained my cybersecurity awareness.
- C4. † The Security Cards were not a source of frustration.

Reporting these

Report counts, not percentages. With nineteen or thirty participants, “31.6%” claims a precision the sample does not have; “6 of 19” is honest and easier to read. A diverging stacked bar beats four pie charts.

5. Part D — Content of the Cards

Tests design goal G1 (vocabulary accessibility), G3 (framework traceability), and requirement R2 (bound the working set).

- D1. The cards used language I could understand without asking for help.
- D2. The examples on the back of the cards helped me decide what a card meant.
- D3. ^R There were too many cards to keep track of.
- D4. ^R Some cards seemed to overlap with each other.
- D5. Seeing that a card referred to a published control, such as “CIS Control 7,” made the card feel more trustworthy.
- D6. If I wanted to learn more about one of these controls after today, I would know where to start.

D3 and D4 are the honest ones

Both are direct quotations of Cycle 1 complaints, reworded as items. If the tiering and the eight-card hand worked, D3 should fall relative to Cycle 1’s open-ended reports. If it does not, say so — a requirement that was not met is a finding, not a failure.

6. Part E — Process and Gameplay

Tests R1 (choices must be contestable) and R3 (basis for feedback on quality). E3 is the single most important item in this instrument.

- E1. The order of the activities was clear to me.
- E2. Having a limited budget made the choices harder in a way that was useful.
- E3. ^R There was usually one obviously correct answer.
- E4. My team disagreed about at least one purchase.
- E5. Having to justify a modifier card out loud made me think more carefully.
- E6. I had a sense of whether our choices were any good.
- E7. We had enough time to finish without rushing.
- E8. ^R The facilitator had to explain the rules more often than I expected.

Reading E3 and E4 together

Cycle 1’s core complaint was that too many answers seemed equally acceptable. E3 is that complaint stated as an item and E4 is its behavioural counterpart. If the budget mechanism did what it was designed to do, E3 falls and E4 rises. If E3 stays high, the budget is too loose — which is a calibration problem with a known fix, not a refutation of the design.

E6 will probably score low. Novices want an answer key and the activity deliberately withholds one. Report it and interpret it rather than treating it as a defect.

7. Part F — Realism and Transfer

A limit worth stating in the paper

Novices are poor judges of whether a security scenario is realistic — they have no basis for comparison, which is why they are in the course. Treat Part F as measuring *perceived relevance and intent to transfer*, not realism. Realism is judged on the facilitator form by someone with domain knowledge, and that is the number to cite if you make a claim about it.

- F1. The organization in the scenario felt like a place that could exist.
- F2. The trade-offs felt like ones a real organization would actually face.
- F3. I can see how this connects to advising an actual client.
- F4. Placing the threat again after buying controls matched how I think risk really behaves.
- F5. ^R This felt more like a puzzle than like a real decision.
- F6. I would use something I learned today if a friend or family member asked me a security question.

8. Part G — Confidence, After Play

Repeat items B1–B5 verbatim, same 0–100 slider, same wording. Label them G1–G5. Do not show participants their earlier answers.

9. Part H — Workload, Features, and Open Responses

9.1. Workload (0–10)

Three items, separating load that comes from the subject from load that comes from how it was presented. The distinction matters: high intrinsic load is expected and fine, high extraneous load is a design defect you can fix.

- H1. The topic itself — threats and security controls — was complex. (*0 = very easy, 10 = very complex*)
- H2. The way the activity was set up and explained made it harder than it needed to be. (*0 = not at all, 10 = very much*)
- H3. The activity helped me understand how threats and controls relate. (*0 = not at all, 10 = very much*)

9.2. Which Parts Were Worth It

Best-worst rather than rating everything, because rating eight features on a five-point scale produces eight fours and tells you nothing.

Of the parts of the activity listed below, which **one** was most useful to you, and which **one** was least useful?

- (a) The four threat cards used to build the scenario
- (b) The Solutions cards describing security measures
- (c) The CIS control references printed on the cards
- (d) The risk matrix
- (e) The limited budget and spending tokens
- (f) The modifier cards and having to justify them
- (g) Placing the threat a second time after buying controls

- (h) Comparing your team's choices with another team's

Scoring best-worst

For each option, score = (times chosen most useful) – (times chosen least useful), divided by the number of respondents. The result runs from –1 to +1 and ranks cleanly. Report the whole ranking, including the negatives — a feature that most participants found least useful is a candidate to cut, and saying so is more credible than reporting only what people liked.

9.3. Open Responses

Three, all specific. Generic prompts such as “any other comments” produce generic answers.

- H4. Which purchase did your team argue about most, and how did you settle it?
 H5. What did you want to buy and could not afford? What would you have told the organization about that gap?
 H6. If you could change one thing about this activity, what would it be?

10. Part I — Facilitator Form

One per facilitator per session, completed the same day. This is where domain judgment lives and where the process data that no participant can report gets captured.

11. Scenario used, budget issued, gameplay variant, number of teams, team sizes.
12. Actual elapsed minutes per phase, and total including debrief.
13. As someone with security background: how realistic was this scenario for the kind of organization it described? (*1–5, plus one line of comment*)
14. Did teams argue substantively about at least one purchase? (*All teams / Most / Some / None*)
15. Roughly how many interventions did you make per table, and at which rung? (*Counts for rungs 1–5 from the facilitator manual*)
16. Did any of the four immediate-intervention situations arise? (*Yes/No per situation; do not record details of disclosures or personal content*)
17. Justification quality, scored 0–3 per team using the manual's rubric.
18. What broke, confused people, or took longer than expected?
19. What would you change before running it again?

On I6

Record only that a situation occurred, never its content. If a participant disclosed a real vulnerability or a scenario touched personal experience, that belongs in your own follow-up, not in a research dataset. Say this explicitly in your IRB protocol.

11. Scoring and Analysis Notes

11.1. Reverse-Scored Items

D3, D4, E3, E8, F5. Recode as $6 - x$ before combining with anything.

11.2. Suggested Groupings

These are provisional. With a first sample you can report item-level results and compute internal consistency; do not present them as validated subscales until you have the data to justify it.

Grouping	Items	What it speaks to
Accessibility	D1, D2, D6, C2	Goal G1, and the Cycle 1 ease-of-use result
Cognitive burden	D3 ^R , D4 ^R , H2	Requirement R2
Contestability	E2, E3 ^R , E4	Requirement R1 — the central design claim
Feedback and rigour	E5, E6	Requirement R3
Traceability	D5, D6	Goal G3
Perceived transfer	F2, F3, F6	Whether it connects to clinic work
Confidence change	G1–G5 minus B1–B5	Paired, per participant

11.3. Analysis

Report counts and medians for ordinal items rather than means alone; with samples this size a mean of 4.42 and a mean of 4.11 are not meaningfully different. Use Wilcoxon signed-rank for the paired confidence change, report an effect size with a confidence interval, and treat the study as powered only for large effects. State the pre/post pairing rate.

11.4. What You Can and Cannot Say

Supported by this instrument	Not supported
Participants found the deck usable and its vocabulary accessible	Participants learned to select appropriate controls
The budget made choices feel contestable	The budget improved decision quality
Confidence on five specific tasks rose after one session	Competence rose, or the change persisted
Participants rated the trade-offs as plausible	The scenarios are realistic (<i>use the facilitator judgment, I3</i>)
Which features participants valued most	Which features contributed most to learning

The right-hand column is answerable, but from the session artifacts — the exported portfolios, matrix deltas, and justification scores — not from a questionnaire.